Pre-Logon VPN for Cloud & Hybrid-Joined Devices — First-Time Off-Network Sign-In & Password Reset

Implementation Solution Guide

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Executive Summary — Pre-Logon VPN for Cloud & Hybrid-Joined Devices: First-Time Off-Network Sign-In & Password Reset

The business problem

Remote-first work breaks two everyday moments that used to "just work" in the office. A **new employee's** laptop is shipped to their home with **no cached credentials**, and they cannot complete their **first Windows sign-in** because the device can't reach a domain controller. And a remote user's **password expires** off-network, but they can't change it at the sign-in screen because the password change also has to reach a domain controller. Both leave people stranded, generate help-desk tickets, and — worst case — result in laptops being couriered around just to log in once.

The solution

This is a complete, self-guided **decision, scenario, and validation** package for **pre-logon VPN** — bringing the network up **before or at** Windows sign-in so a domain controller is in reach at exactly the moment it's needed. It does the part teams actually get wrong: it **chooses the right approach** for your fleet (Windows Always On VPN **device tunnel** vs. Cisco Secure Client **Start Before Logon / PLAP**), explains the **two hard scenarios by device join type**, and **validates** them end to end — proving a never-logged-on user can first-sign-in and an expired-password user can reset, both off the network. It uses a proven method: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand**, with explicit pass/fail gates.

What you get

- **A defensible decision** — a device-tunnel-vs-SBL/PLAP matrix by join type, OS SKU, existing VPN vendor, and support model, so you pick once and pick right.
- **The two scenarios covered** — first-time off-network logon and off-network password change, explained by join type (why it's a **hybrid/domain-joined** problem, and why **Entra-joined** devices largely avoid it), with SSPR positioned as the cloud complement.
- **Five PowerShell scripts** — fleet join-type + readiness discovery, on-device validation (join type, pre-logon path, DC reachability), a coverage report, safe remediation (preview-by-default), and a segmentation-group builder.
- **Requirements, testing, troubleshooting, and best-practice guides** so a competent admin can decide, validate, and operate it.

Business outcomes

- **New hires sign in on day one, from anywhere** — no couriered laptops, no "log in at the office first" workaround.

• **Expired passwords self-resolve off-network** — the change reaches a domain controller through the pre-logon tunnel, cutting lockout tickets.

• **No wasted deployment** — clear guidance that cloud-only Entra-joined devices usually need neither approach, so you don't ship a tunnel where none is required.

• **Proven before it matters** — both scenarios validated on a pilot with explicit gates, and a documented break-glass fallback for the day something breaks.

Scope and boundary

This package is the **decision + scenario + validation** layer. The **step-by-step builds** for the device tunnel and for SBL/PLAP are **separate companion products**, referenced here by their exact object names so the decision, the build, and the validation stay coherent — this solution does not recreate them. VPN gateway and vendor client internals follow those products and the **vendor's official documentation**.

Bottom line

The two remote-work sign-in problems — first logon with no cached credentials, and password reset off the network — solved by choosing the right pre-logon approach and proving it works, at a fraction of the cost and time of a hands-on consulting engagement.

Solution Overview — Pre-Logon VPN for Cloud & Hybrid-Joined Devices: First-Time Off-Network Sign-In & Password Reset

What this solution is

This solution is the **decision, scenario, and validation layer** for pre-logon VPN. It answers the two questions that a remote-first fleet actually gets stuck on:

1. **A brand-new employee's laptop is off the corporate network and has *no cached credentials* — how do they complete their *first* Windows sign-in?**

1. **A remote user's password has *expired* while off-network — how do they *change* it at the logon/lock screen when the password change has to reach a domain controller?**

Both problems have the same root cause: the network is not up *before* Windows sign-in, so the device cannot reach a **domain controller (DC)** at the moment it needs one. **Pre-logon VPN** solves both by bringing the tunnel up **before or at** the Windows credential prompt.

This solution does **not** re-teach the two deployment builds that create a pre-logon tunnel. Those are **separate products** (see below). This product tells you **which approach to choose**, **how the two scenarios actually behave** by device join type, and **how to validate end to end** that a never-logged-on user and an expired-password user can both get in from off the network.

What this solution is NOT (and where to get it)

• It does **not** contain the step-by-step **Always On VPN device-tunnel** build. That is the

companion product *Always On VPN — Device Tunnel (Pre-Logon) via Intune*, which creates the device-tunnel profile named `AOVPN – Device Tunnel – ProfileXML – Windows` (plus its trusted-root and machine-certificate profiles).

• It does **not** contain the step-by-step **Start Before Logon (SBL/PLAP)** build. That is the

companion product *Pre-Logon VPN — Cisco Secure Client (SBL/PLAP) via Intune*, which creates the Win32 app named `Cisco Secure Client – Pre-Logon (SBL) – Win32` (plus its SBL profile).

This product **references those objects by name** so the three pieces stay coherent, but it does not duplicate their builds. If you have not deployed either approach yet, you run this decision guide **first** to pick the right one, then implement the chosen companion product, then return here to validate the two scenarios.

The two approaches you are choosing between

Pre-logon connectivity comes from one of two mechanisms. This solution's job is to help you pick the right one for your fleet — not to reconfigure them.

Approach A — Windows Always On VPN device tunnel

• **What it is:** A machine-level IKEv2 VPN that Windows brings up **automatically at boot, before

any user signs in**, authenticated by a** machine certificate** (no user interaction).

• **Hard requirements (facts):** the device must be **domain-joined or Entra hybrid-joined**, run

Windows Enterprise or Education SKU, hold a **machine certificate** the VPN gateway trusts, and the gateway must support **IKEv2**.

• **Best when:** you want a *silent, always-available* DC path for every managed Windows device with

no user action, and you own the PKI + an IKEv2-capable gateway.

• **Object it deploys (companion product):** `AOVPN – Device Tunnel – ProfileXML – Windows`.

Approach B — Cisco Secure Client Start Before Logon (SBL / PLAP)

• **What it is:** A **Windows credential provider (PLAP)** that lets the user bring up the Cisco

Secure Client VPN **at the Windows logon screen**, *before* submitting their Windows credentials.

• **Hard requirements (facts):** Windows only; the Secure Client with the **SBL/PLAP** module

installed; a reachable Cisco head-end. It is **user-initiated at the logon screen**, not silent.

• **Best when:** you already run Cisco Secure Client (AnyConnect), you cannot meet the device-tunnel

SKU/PKI/IKEv2 requirements, or you want the user to choose the network before signing in.

• **Object it deploys (companion product):** `Cisco Secure Client – Pre-Logon (SBL) – Win32`.

The decision matrix (summary)

Factor	Device tunnel (A)	SBL / PLAP (B)
Join type supported	Domain-joined **or** Entra hybrid-joined	Domain-joined **or** hybrid-joined (Windows)
OS SKU	**Enterprise / Education required**	Any Windows Pro/Enterprise (per Cisco support)
Auth at pre-logon	**Machine certificate (silent)**	User brings up VPN at logon screen
Tunnel protocol	**IKEv2**	Cisco Secure Client (per head-end)
User action needed	**None** (auto at boot)	User connects at the logon screen
Existing VPN vendor fit	Native Windows / any IKEv2 gateway	You already run Cisco Secure Client
Entra-joined (cloud-only) devices	Usually **not needed** (see below)	Usually **not needed** (see below)

The full matrix, with a step-by-step "choose the approach" procedure, is in `03-implementation.md`.

Why join type is the whole story

The first-logon / DC-reachability problem is fundamentally a **hybrid-joined / domain-joined** problem — because those devices authenticate the user against **Active Directory**, which lives on a **domain controller** the device must reach.

- **Entra-joined (cloud-only) devices** authenticate the user against **Entra ID in the cloud**, not

against an on-prem DC. A first sign-in works over ordinary internet with no pre-logon tunnel, and a password change is a **cloud** operation. These devices **largely avoid** the classic first-logon and off-network password-change problems this product solves. Do not deploy a pre-logon tunnel to solve a problem an Entra-joined device does not have.

- **Entra hybrid-joined and domain-joined devices** still need a DC for the *interactive* logon of a

user who has **never signed in on that device** (no cached credentials) and for an **AD password change**. These are the devices pre-logon VPN is for.

Getting this right up front avoids the two most common mistakes: deploying a pre-logon tunnel to cloud-only devices that never needed it, and expecting SSPR (a cloud feature) to fix an on-prem AD password change on a hybrid device.

Scenario 1 — First-time off-network logon

A new hire's hybrid-joined laptop is shipped to their home. They have **never logged on** to that device, so Windows has **no cached credential** for them. Without a DC in reach at the logon screen, the *first* interactive logon cannot be validated and fails.

- **Cached credentials** only help a user who has **already** signed in on that device — Windows

caches a verifier after a successful DC logon. A never-logged-on user has **nothing cached**, so they **require** a live DC path for the first logon.

• **Pre-logon VPN provides that path.** Device tunnel (A) brings the DC into reach automatically before the logon screen; SBL/PLAP (B) lets the user bring it up at the logon screen. Either way, the first interactive logon can now reach a DC and succeed.

• **Validation** (see `05-testing.md`, T1) proves a freshly provisioned device with a never-logged-on user completes first logon off-network via the pre-logon tunnel.

Scenario 2 — Off-network password change / reset

A remote user's password has **expired**. Windows prompts them to change it at the logon/lock screen — but an **Active Directory password change** must be written to a **domain controller**. Off the network with no pre-logon tunnel, the change cannot reach a DC and fails.

• **The tunnel must be up so the change reaches a DC.** With device tunnel (A) the machine tunnel is already up at the lock screen; with SBL/PLAP (B) the user brings the tunnel up first, then changes the password. The Kerberos/AD password change then completes against the DC.

• **SSPR (Entra self-service password reset) is the cloud-native complement.** For **Entra-joined** users, SSPR changes the cloud password and there is no on-prem DC dependency. For **hybrid** users, SSPR with **password writeback** can reset the *directory* password, but the **Windows on-prem credential at the lock screen still expects an AD change that reaches a DC** — SSPR does **not** remove the need for a DC path for the local interactive password change on a hybrid/domain-joined device. Treat SSPR as complementary, not a replacement, and validate the actual behavior (T2).

Break-glass and fallback

Pre-logon VPN is infrastructure, and infrastructure fails. This solution documents the fallbacks:

• **Local administrator account** on the device for recovery (kept per your security policy).

• **Temporary Access Pass (TAP)** in Entra for passwordless first sign-in / recovery where the flow is cloud-capable.

• **What to do if pre-logon VPN fails** — bring the device on-network once (office/known network), use SBL/PLAP as a manual fallback where a silent device tunnel did not come up, or use SSPR/TAP for the cloud-capable paths. Full guidance is in `06-troubleshooting.md`.

Benefits

• **The right approach, chosen once.** A defensible device-tunnel-vs-SBL decision by join type, OS SKU, existing vendor, and support model — not a coin toss.

• **The two hard scenarios actually covered.** First-time off-network logon and off-network password change, explained by join type and validated end to end.

• **No cloud-only over-deployment.** Clear guidance that Entra-joined devices usually need neither approach, so you don't ship a tunnel where none is required.

• **Coherent with the deployment products.** Objects are referenced by their exact companion-product

names, so the decision, the build, and the validation all line up.

• **Reportable readiness.** The included scripts report join types across the fleet and which devices already have a pre-logon path, so you know your coverage before an employee is stranded.

Limitations and boundaries

• **This is the decision + scenario + validation layer, not a deployment build.** The device-tunnel and SBL/PLAP step-by-step builds live in their companion products and are referenced, not repeated.

• **Vendor and OS internals are deferred.** IKEv2/ProfileXML internals and the Cisco head-end / SBL module internals follow the companion products and the **vendor's official documentation**.

• **Active Directory and PKI are assumed.** A reachable DC (through the tunnel) and, for the device tunnel, a machine-certificate PKI path are prerequisites, not deliverables.

The standard model

This solution follows the category-standard model, with explicit pass/fail gates — but framed around **scenario validation** rather than profile creation:

Assess → **Configure** → **Pilot** → **Validate** → **Expand**

1. **Assess.** Inventory device join types and existing pre-logon capability; confirm prerequisites.
2. **Configure.** *Choose* the approach with the decision matrix and confirm the chosen companion product's objects exist; segment the pilot users.
1. **Pilot.** Put a representative pilot cohort (hybrid + domain-joined + one Entra-joined control) into `PILOT – Pre-Logon First-Logon Users`.
1. **Validate.** Run the two decisive scenario tests (first-logon, password-change) end to end.
2. **Expand.** Widen to production, monitor readiness coverage, keep the break-glass fallback ready.

Proceed to the **Requirements guide** (`02-requirements.md`) to confirm you are ready to decide and validate.

Requirements — Pre-Logon VPN: First-Time Off-Network Sign-In & Password Reset

Confirm **every** requirement below before you decide and validate. Each entry follows the same shape: **Description / Why required / How to verify / How to implement if missing.** The category-level prerequisites (Entra tenant, Intune, admin roles, pilot group, change control) are in `global-requirements-network.md`; the items here are specific to choosing a pre-logon approach and validating the first-logon and off-network password-change scenarios.

> ***Scope boundary:*** *this product is the* ***decision + scenario + validation*** *layer. The step-by-step device-tunnel and SBL/PLAP builds are* ***separate companion products*** *and are referenced by their object names — this guide does* ***not*** *re-teach them and does not fabricate vendor menus. Where a*

requirement says "from the companion product," it means the object is created there, not here.

R1. A chosen (or deployed) pre-logon approach

Description: Either a working **Always On VPN device tunnel** or a working **Cisco Secure Client Start Before Logon (SBL/PLAP)** deployment — or a firm decision to deploy one of them, made with the decision matrix in `03-implementation.md`. **Why required:** This product validates that a pre-logon path exists and works for the two scenarios. It cannot validate a capability the fleet does not have. **How to verify:** On a representative device, confirm a pre-logon tunnel exists — `Get-VpnConnection -AllUserConnection` shows a device tunnel, **or** the SBL/PLAP credential provider (`vpnplap64.dll`) is present. The included `PREVPN-Validation.ps1` checks both. **How to implement if missing:** Run the decision matrix (`03-implementation.md`, Phase 2) to choose, then deploy the chosen **companion product** — *Always On VPN — Device Tunnel (Pre-Logon) via Intune* (creates `AOVPN – Device Tunnel – ProfileXML – Windows`) **or** *Pre-Logon VPN — Cisco Secure Client (SBL/PLAP) via Intune* (creates `Cisco Secure Client – Pre-Logon (SBL) – Win32`).

R2. Entra ID security groups (pilot and production)

Description: A pilot group `PILOT – Pre-Logon First-Logon Users` and one or more production target groups (e.g. `PROD – Pre-Logon First-Logon Users`) containing the users/devices in scope. **Why required:** Segmentation and reporting target Entra groups. The scenario validation is run against a small, named pilot cohort before expanding. **How to verify:** In the **Microsoft Entra admin center** (`https://entra.microsoft.com`) → **Groups > All groups**, confirm `PILOT – Pre-Logon First-Logon Users` exists with the intended members. **How to implement if missing:** Entra → **Groups > New group > Security**; name it exactly `PILOT – Pre-Logon First-Logon Users`; add 3–10 representative members. The included `PREVPN-Set-ScopeTargeting.ps1` creates and populates these groups for you.

R3. A device join-type inventory

Description: A current inventory of how your Windows devices are joined — **domain-joined**, **Entra hybrid-joined**, or **Entra-joined (cloud-only)** — because the pre-logon decision and the scenario behavior differ entirely by join type. **Why required:** The first-logon / DC-reachability problem is a **hybrid/domain-joined** concern; Entra-joined devices largely avoid it. Choosing an approach or scoping a pilot without the join-type mix leads to over-deploying to cloud-only devices. **How to verify:** Run `PREVPN-Discovery.ps1` for a fleet join-type report, or spot-check a device with `dsregcmd /status` (look at `AzureAdJoined`, `DomainJoined`, `EnterpriseJoined`). **How to implement if missing:** Produce the inventory with `PREVPN-Discovery.ps1` before deciding. Record the counts of each join type; this drives the decision matrix.

R4. A reachable domain controller through the tunnel (hybrid/domain-joined)

Description: At least one **Active Directory domain controller** that a pre-logon device can reach **once the tunnel is up** — for the first interactive logon and for AD password changes. **Why required:** Both scenarios depend on the device reaching a DC. A pre-logon tunnel that comes up but has **no route to a**

DC solves nothing. **How to verify:** With the tunnel up on a test device, `nltest /dsgetdc:<domain>` locates a DC, or `Test-NetConnection <dc-fqdn> -Port 389` succeeds. `PREVPN-Validation.ps1` performs a best-effort DC reachability check. **How to implement if missing:** Ensure the VPN gateway's routes/split-tunnel include the DC subnets and that DNS resolves the domain through the tunnel (companion-product/network task). Confirm firewall allows the AD ports (DNS 53, Kerberos 88, LDAP 389, etc.) from the VPN client range to the DCs.

R5. Machine/device certificate path (only if choosing the device tunnel)

Description: A working PKI that issues each device a **machine certificate** the VPN gateway trusts — delivered by the companion device-tunnel product (SCEP/PKCS via NDES, Intune Cloud PKI, or a third-party CA connector). **Why required:** The device tunnel authenticates **silently with a machine certificate** over IKEv2. No machine cert → no silent pre-logon tunnel. **How to verify:** On a device, a valid certificate is present in **LocalMachine\My** and chains to a CA the gateway trusts (`PREVPN-Validation.ps1` checks for a LocalMachine\My cert). **How to implement if missing:** Deploy the machine-cert profile from the companion device-tunnel product (`AOVPN – Device Tunnel – Machine Cert (SCEP) – Windows`) and the trusted-root profile. **Skip R5 entirely if you choose SBL/PLAP** (Approach B), which does not require a machine certificate.

R6. Windows Enterprise / Education SKU (only if choosing the device tunnel)

Description: The **Windows Enterprise or Education** edition on devices that will run the Always On VPN **device tunnel**. **Why required:** The device tunnel is a **hard OS-SKU requirement** — it is supported only on Enterprise/Education, not on Windows Pro/Home. **How to verify:** `Get-ComputerInfo | Select WindowsProductName`, or **Settings > System > About**, shows Enterprise or Education. **How to implement if missing:** Move affected devices to a qualifying SKU, **or** choose **SBL/PLAP** (Approach B), which does not have this SKU restriction.

R7. Entra SSPR (optional — cloud password complement)

Description: Entra self-service password reset (SSPR), optionally with **password writeback** for hybrid users, as the cloud-native complement to the on-prem password-change scenario. **Why required:** SSPR resolves the password path for **Entra-joined** users and, with writeback, the directory password for hybrid users — reducing (but not eliminating) DC-path dependence at the lock screen. Understanding where it does and does not help prevents wrong expectations. **How to verify:** Entra → **Protection > Password reset** shows SSPR enabled and (for hybrid) on-premises writeback configured and healthy. **How to implement if missing:** Enable SSPR (Entra → **Password reset**) and, for hybrid, configure password writeback via Entra Connect. **Note:** SSPR does **not** remove the need for a DC path for a hybrid/domain-joined device's local interactive password change — validate the real behavior in T2.

R8. Break-glass / recovery accounts

Description: A documented fallback: a **local administrator** account for on-device recovery and a **Temporary Access Pass (TAP)** policy in Entra for passwordless first sign-in / recovery where the flow is cloud-capable. **Why required:** Pre-logon VPN can fail (bad cert, gateway down, DC unreachable). A break-glass path prevents a stranded new hire or a locked-out remote user. **How to verify:** A local admin

account exists per policy; Entra → **Authentication methods > Temporary Access Pass** is enabled for the appropriate users. **How to implement if missing:** Establish the local-admin recovery policy; enable TAP in Entra and document who may issue one. Capture the fallback procedure in your runbook.

R9. Administrative roles

Description: Entra rights to create/populate groups (Groups Administrator or better) and read device/join information, plus Intune read access to confirm the companion objects. **Why required:** Segmenting the pilot cohort, reading join types, and confirming the pre-logon objects require appropriate roles. **How to verify:** In Entra you can create groups and read devices; in Intune you can view apps and configuration profiles. **How to implement if missing:** Have a Global Administrator assign **Groups Administrator** and **Intune Administrator** (least-privilege) to the operator.

R10. Microsoft Graph PowerShell SDK (to run the included scripts)

Description: The **Microsoft.Graph** PowerShell SDK on the operator workstation, used by the discovery, report, remediation, and scope-targeting scripts. (`PREVPN-Validation.ps1` runs locally on a device and needs no Graph.) **Why required:** The scripts call Microsoft Graph (Entra/Intune) to report join types, pre-logon capability presence, and group membership. **How to verify:** `Get-Module Microsoft.Graph -ListAvailable` returns a version. **How to implement if missing:** `Install-Module Microsoft.Graph -Scope CurrentUser`. Each script's header lists the exact delegated scopes it uses.

R11. Change control and rollback (category requirement, restated)

Description: An approved change window and a written rollback for any pre-logon change you make as a result of this decision (revert the chosen approach's assignment, fall back to on-network onboarding). **Why required:** Pre-logon connectivity affects how real users sign in; any change must be reversible and must never strand a remote user. **How to verify:** A rollback and comms plan exist and are approved before you change targeting. **How to implement if missing:** Document the rollback (exclude the group from the companion product's assignment; use the break-glass path from R8) before you expand.

When all requirements pass, proceed to `03-implementation.md` to choose the approach and validate the two scenarios.

Implementation — Pre-Logon VPN: Choose the Approach & Validate the Two Scenarios

This is the ordered procedure. Follow the phases in order: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand.** Because this product is the **decision + scenario + validation** layer, the phases are framed around **choosing an approach** and **proving the two scenarios**, not around building VPN profiles (those builds live in the companion products). Use the **exact object names** shown — the tests, troubleshooting, and scripts reference them verbatim.

> ***Scope boundary:*** *the device-tunnel build (*`AOVPN – Device Tunnel – ProfileXML – Windows`*) and the SBL/PLAP build (*`Cisco Secure Client – Pre-Logon (SBL) – Win32`*) are* ***created in their companion products*** *and are* ***deferred to those products and the vendor's***

> ***official documentation**. This guide chooses between them and validates the outcome; it does not re-create them.*
>
> ***Government / GCC High note:** the portal URLs below are Commercial. In US sovereign clouds use the `*.us` equivalents — Intune/Endpoint at `https://endpoint.microsoft.us`, Entra at `https://entra.microsoft.us`, and sign-in at `login.microsoftonline.us` — and **verify each step for GCC High**, as feature availability and portal paths can differ. Do not assume Commercial steps map one-to-one; confirm against Microsoft US Gov documentation.*

Object naming convention (use verbatim)

Object	Name	Created where
Pilot group	`PILOT – Pre-Logon First-Logon Users`	This product (`PREVPN-Set-ScopeTargeting.ps1`)
Production group	`PROD – Pre-Logon First-Logon Users`	This product (`PREVPN-Set-ScopeTargeting.ps1`)
Device-tunnel profile (referenced)	`AOVPN – Device Tunnel – ProfileXML – Windows`	Companion device-tunnel product
SBL/PLAP app (referenced)	`Cisco Secure Client – Pre-Logon (SBL) – Win32`	Companion SBL/PLAP product

Phase 1 — Assess

1. Confirm every item in `02-requirements.md` passes.
2. Build the **join-type inventory** and pre-logon readiness baseline:

`powershell -ExecutionPolicy Bypass -File .\scripts\PREVPN-Discovery.ps1` Review the readiness table: counts by join type (hybrid / domain-joined / Entra-joined), whether a device-tunnel profile **or** the SBL/PLAP app is present, and pilot/prod group membership.

1. Note which devices are **hybrid/domain-joined** (the ones the two scenarios apply to) versus

Entra-joined (which usually need neither approach).

Gate A (Assess): You have the join-type mix, you know which devices are in scope for pre-logon, and all requirements pass. If the fleet is overwhelmingly Entra-joined, confirm you actually need pre-logon at all before proceeding.

Phase 2 — Configure (choose the approach)

2.0 Create the segmentation groups

Use the script (recommended) or the portal.

- **Script:** `powershell -ExecutionPolicy Bypass -File .\scripts\PREVPN-Set-ScopeTargeting.ps1`

(creates `PILOT – Pre-Logon First-Logon Users` and, with `-CreateProd`, `PROD – Pre-Logon First-Logon Users`, and adds the members you supply).

- **Portal:** Entra admin center → **Groups > New group > Security** → name

`PILOT – Pre-Logon First-Logon Users` → add pilot members → **Create**. Repeat for the prod group.

2.1 Run the decision matrix — pick device tunnel or SBL/PLAP

Work down the matrix. The **first row that forces a choice wins**; use the later rows to confirm.

#	Question	If this is true →	Approach
1	Are the in-scope devices **Entra-joined (cloud-only)**?	They authenticate against Entra in the cloud; a first logon and password change work without an on-prem DC. **Do not deploy a pre-logon tunnel** — use SSPR/TAP for the cloud paths.	**Neither**
2	Are devices **domain-joined / hybrid-joined**?	The scenarios apply — continue to row 3.	(continue)
3	Are all in-scope devices **Windows Enterprise / Education** with a **machine-cert PKI** and an **IKEv2** gateway?	You can run a **silent, no-user-action** machine tunnel at boot.	**Device tunnel (A)**
4	Do you **already run Cisco Secure Client**, or **cannot** meet the Enterprise/Education + machine-cert + IKEv2 bar?	Use the **logon-screen** VPN the user brings up before signing in.	**SBL / PLAP (B)**
5	Do you want the **user to choose** the network before logon, or need a manual fallback?	SBL/PLAP is user-initiated at the logon screen.	**SBL / PLAP (B)**
6	Support model: fully hands-off vs. help-desk-guided first connect?	Hands-off → device tunnel (A); guided/manual → SBL/PLAP (B).	(tie-break)

Record the decision (approach, and why) in your runbook. If devices split across categories, you may run **both** (device tunnel for the Enterprise/Education + PKI cohort, SBL/PLAP for the rest) — segment each cohort with the pilot/prod groups.

2.2 Confirm the chosen approach's objects exist

You do **not** build the tunnel here — you confirm the companion product deployed it.

- **If you chose the device tunnel (A):** confirm `AOVPN – Device Tunnel – ProfileXML – Windows`

exists and is assigned (deploy it via the companion device-tunnel product if not). `PREVPN-Discovery.ps1` flags whether a device-tunnel profile is present.

- **If you chose SBL/PLAP (B):** confirm `Cisco Secure Client – Pre-Logon (SBL) – Win32` exists and is

assigned (deploy it via the companion SBL/PLAP product if not). `PREVPN-Discovery.ps1` flags whether the SBL app is present.

Gate B (Configure): The approach is chosen and recorded, the segmentation groups exist, and the chosen approach's object is present and assigned. Re-run `PREVPN-Discovery.ps1` — the fleet should now show a pre-logon path for the in-scope devices.

Phase 3 — Pilot

1. Put a **representative** cohort into `PILOT – Pre-Logon First-Logon Users`:

- at least one **hybrid-joined** device,
- at least one **domain-joined** device (if you have any),
- and one **Entra-joined** device as a **control** (to confirm it does *not* need pre-logon).

1. Ensure the chosen approach's object (device-tunnel profile or SBL app) is assigned so the pilot

devices actually have the pre-logon capability.

1. On a pilot device, confirm the capability landed:

`powershell -ExecutionPolicy Bypass -File .\scripts\PREVPN-Validation.ps1` It reports the join type, whether a pre-logon tunnel exists, the machine cert (if device tunnel), and a best-effort DC reachability check.

Gate C (Pilot): Every in-scope pilot device shows a pre-logon path present and a DC reachable through it (`PREVPN-Validation.ps1` = PASS). The Entra-joined control shows *no* pre-logon path and still signs in — confirming it did not need one. If a pilot device fails, go to `06-troubleshooting.md`.

Phase 4 — Validate (the two decisive scenarios)

These are the tests this product exists for. Run both on the pilot before expanding. Full pass/fail detail is in `05-testing.md` (T1 and T2).

1. **Scenario 1 — First-time off-network logon.** Take a **freshly provisioned** hybrid/domain-joined

device with a **user who has never logged on to it**, keep it **off the corporate network**, and confirm the **first interactive logon succeeds** because the pre-logon tunnel brought a DC into reach (device tunnel auto-up at the logon screen, or the user brings up SBL/PLAP first).

1. **Scenario 2 — Off-network password change.** Take a user whose **password is expired** (or force a

change), off-network, and confirm they can **change the password at the logon/lock screen** and sign in — because the tunnel carried the AD password change to a DC.

1. Run the fleet readiness roll-up to see pilot coverage:

```
powershell -ExecutionPolicy Bypass -File .\scripts\PREVPN-Report.ps1
```

Gate D (Validate): On the pilot, **both** scenarios PASS on every in-scope join type, and the Entra-joined control confirms the cloud paths (ordinary logon; SSPR/TAP for password). All Testing gates are green. Only then expand.

Phase 5 — Expand

1. Add `PROD – Pre-Logon First-Logon Users` to the chosen approach's assignment (via the companion

product), keeping the pilot group assigned. Ensure only **hybrid/domain-joined** devices are targeted — do not push a pre-logon tunnel to Entra-joined devices that don't need it.

1. Expand in waves for large fleets; watch `PREVPN-Report.ps1` between waves for coverage of the

pre-logon path across in-scope devices.

1. Remediate stragglers with `PREVPN-Remediation.ps1` (defaults to **-WhatIf** preview — it adds

devices/users lacking a pre-logon path to the pilot group for the chosen approach and prints guidance; it does **not** create VPN profiles — that is the companion products' job).

1. Keep the **break-glass fallback** (R8: local admin, TAP, on-network onboarding) documented and

ready for any device where pre-logon VPN fails.

Rollback (any phase): Exclude the target group from the chosen approach's assignment (in the companion product) and fall back to on-network first onboarding / SSPR / TAP per R8. No VPN profile is created by this product, so rollback here is purely a targeting/segmentation change.

Gate E (Expand): In-scope coverage meets your target (e.g. ≥ 95% of hybrid/domain-joined devices have a validated pre-logon path); Entra-joined devices are correctly excluded; both scenarios remain green on spot-checks. The decision is implemented and validated.

Testing & Validation — Pre-Logon VPN: First-Logon & Off-Network Password Reset

Run these tests on the **pilot** before expanding, and spot-check during expansion. Each test has an explicit **pass/fail gate**. A phase is only "done" when its gate is **PASS**. Object names match the Implementation

guide verbatim. The two decisive scenario tests are **T1 (first-time off-network logon)** and **T2 (off-network password change)** — this product exists to prove them.

> ***Scope note:*** *these tests validate the* ***outcome*** *— that a never-logged-on user and an expired-password user can get in from off the network — and that the pre-logon path and DC reachability are present. Building the tunnel itself is done in the companion products; deep VPN gateway / head-end diagnostics are deferred to those products and the vendor's documentation.*

Test matrix overview

#	Test	Where	Pass gate
T1	**First-time off-network logon**	Freshly provisioned pilot device	Never-logged-on user completes first logon off-network
T2	**Off-network password change**	Pilot device, expired password	User changes password at logon/lock screen and signs in
T3	Pre-logon path present	On-device validation script	Tunnel (device tunnel or SBL/PLAP) present
T4	DC reachable through the tunnel	Pilot device	A DC is located/reachable once the tunnel is up
T5	Machine cert present (device tunnel only)	Pilot device	Valid cert in LocalMachine\My, chains to trusted CA
T6	Entra-joined control behaves correctly	Entra-joined device	Signs in with no pre-logon tunnel; SSPR/TAP cover password
T7	Fleet readiness roll-up	Report script	In-scope coverage ≥ target
T8	Break-glass / fallback	Test device	Local admin / TAP / on-network path recovers a failed device

T1 — First-time off-network logon (the decisive test #1)

Steps: Take a **freshly provisioned** hybrid-joined (or domain-joined) device that a specific test user has **never signed in to** (no cached credential for that user). Keep the device **off the corporate network** (e.g. home Wi-Fi / hotspot). Attempt the user's **first interactive Windows logon**:

- **Device tunnel (A):** the machine tunnel should already be up at the logon screen (auto at boot); submit the credentials.
- **SBL / PLAP (B):** at the logon screen, use the network/VPN option to bring up `Cisco Secure Client – Pre-Logon (SBL) – Win32` first, then submit the credentials.

PASS: The first interactive logon **succeeds** off-network — the DC was reachable through the pre-logon tunnel and

validated the never-before-cached user. **FAIL:** Logon rejected / "no logon servers available" → see Troubleshooting §First logon still fails (tunnel not up before credential submit, DC not reachable, or wrong approach for the join type).

T2 — Off-network password change (the decisive test #2)

Steps: For a pilot user, **expire the password** (or set "must change at next logon"). Off-network, have the user reach the **logon/lock screen** and attempt the **password change**:

- **Device tunnel (A):** the machine tunnel is up at the lock screen; change the password.
- **SBL / PLAP (B):** bring up the pre-logon VPN first, then change the password.

PASS: The password change **completes** and the user signs in — the AD/Kerberos password change reached a DC through the tunnel. **FAIL:** "Unable to change password" / change not accepted → see Troubleshooting §Password change fails (no DC path, clock skew). Note that **SSPR alone does not** satisfy a hybrid/domain-joined device's local interactive change — confirm the DC path.

T3 — Pre-logon path present

Steps: On a pilot device run `PREVPN-Validation.ps1`. **PASS:** The script reports a pre-logon tunnel present — a **device tunnel** (`Get-VpnConnection -AllUserConnection`) **or** the **SBL/PLAP** credential provider (`vpnplap64.dll`) — matching the chosen approach. **FAIL:** No pre-logon capability found → the chosen approach's object is not deployed/assigned; fix via the companion product, then re-run.

T4 — DC reachable through the tunnel

Steps: With the tunnel up on a pilot device, run `PREVPN-Validation.ps1` (it performs a best-effort DC check), or manually `nltest /dsgetdc:<domain>` / `Test-NetConnection <dc-fqdn> -Port 389`. **PASS:** A domain controller is located and reachable through the tunnel. **FAIL:** No DC found / unreachable → see Troubleshooting §First logon still fails (routes/DNS/firewall for the DC subnets through the tunnel).

T5 — Machine certificate present (device-tunnel deployments only)

Steps: On a device-tunnel pilot device, confirm a valid machine certificate is in **LocalMachine\My** and chains to a CA the gateway trusts (`PREVPN-Validation.ps1 -RequireCertificate`). **PASS:** Certificate present, not expired, correct chain. **FAIL:** No/invalid cert → the silent tunnel cannot authenticate; fix the machine-cert profile in the companion device-tunnel product. (Not applicable to SBL/PLAP.)

T6 — Entra-joined control behaves correctly

Steps: On the **Entra-joined** control device, confirm a first sign-in and (if applicable) a password change work **without** any pre-logon tunnel; confirm SSPR/TAP cover the cloud password path. **PASS:** The Entra-joined device signs in over ordinary internet with **no** pre-logon path present — confirming cloud-only devices do not need this deployment. **FAIL:** If it appears to need a pre-logon tunnel, re-check the join type (`dsregcmd /status`) — it may actually be hybrid-joined, not Entra-joined.

T7 — Fleet readiness roll-up

Steps: Run `PREVPN-Report.ps1`; review counts by join type and the **coverage %** of in-scope (hybrid/domain-joined) devices that have a pre-logon path. **PASS (pilot):** 100% of in-scope pilot devices have a validated pre-logon path. **PASS (expand):** Coverage ≥ your target (e.g. ≥ 95%) with remaining gaps triaged. **FAIL:** Coverage below target → remediate with `PREVPN-Remediation.ps1` (adds lacking devices/users to the pilot group for the chosen approach) before widening.

T8 — Break-glass / fallback verification

Steps: On a disposable test device, simulate a pre-logon failure (e.g. gateway unreachable) and confirm a recovery path works: **local administrator** logon, **TAP** for a cloud-capable first sign-in, or bringing the device **on-network** once. **PASS:** At least one break-glass path recovers the device without stranding the user. **FAIL:** No fallback works → establish the R8 break-glass path before expanding.

Overall exit criteria (before Expand)

T1 and T2 PASS on every in-scope join type, T3–T5 PASS on the pilot, T6 confirms the Entra-joined control needs no tunnel, T7 shows 100% in-scope pilot coverage, and T8 confirms a working break-glass path. Record results against each gate; only then proceed to Phase 5 (Expand).

Troubleshooting — Pre-Logon VPN: First-Logon & Off-Network Password Reset

Common failures and fixes, grouped by symptom. Object names match the Implementation guide. Work top-down within a section — the first cause listed is the most common.

> ***Scope boundary:*** *the pre-logon tunnel itself is built in the* ***companion products*** *(device tunnel / SBL-PLAP) and gateway/head-end internals are deferred to those products and the vendor's official documentation. This guide resolves the* ***decision + scenario*** *problems — first-logon, off-network password change, join-type confusion, and DC reachability — and tells you when a failure points back at the companion build or the network.*

§First logon still fails — the never-logged-on user can't sign in off-network

Symptom: A freshly provisioned device with a user who has never signed in shows "There are currently no logon servers available to service the logon request" (or similar) off-network.

1. **Tunnel not up *before* credentials are submitted.** The DC path must exist **at** the credential

prompt. For the **device tunnel**, confirm it comes up automatically at boot/logon screen (`Get-VpnConnection -AllUserConnection` shows it; it must be a **device/machine** tunnel, not a user tunnel). For **SBL/PLAP**, the user must bring up the pre-logon VPN at the logon screen **first**, then submit credentials — if they type credentials before connecting, there is no DC path yet.

1. **DC not reachable through the tunnel.** Even with the tunnel up, the DC subnets/DNS may not route

through it. Run `PREVPN-Validation.ps1` (DC check) or `nltest /dsgetdc:<domain>` with the tunnel up. Fix routes/split-tunnel and DNS so the DC is reachable, and open the AD ports (DNS 53, Kerberos 88, LDAP 389, etc.) from the VPN client range to the DCs (network/companion task).

1. **Wrong approach for the join type.** If the device is **Entra-joined (cloud-only)**, it does **not**

use an on-prem DC and does not need a pre-logon tunnel — a "failure" here usually means the device was mis-identified. Confirm with `dsregcmd /status` (`AzureAdJoined=YES` + `DomainJoined=NO` = Entra-joined). If it is genuinely hybrid/domain-joined, ensure the chosen approach's object is deployed and assigned.

1. **(Device tunnel) machine cert missing/invalid.** No valid **LocalMachine\My** cert that the gateway

trusts → the silent tunnel never authenticates, so no DC path at logon. Fix the machine-cert profile in the companion device-tunnel product (`PREVPN-Validation.ps1 -RequireCertificate` confirms).

1. **(Device tunnel) OS SKU not Enterprise/Education.** The device tunnel is unsupported on Windows

Pro/Home. Confirm the SKU (R6) or switch that device to SBL/PLAP.

1. **Cached-credential misunderstanding.** Cached credentials only help a user who has **already**

signed in on that device. A never-logged-on user has nothing cached and **requires** a live DC path — this is expected, not a bug; the fix is the pre-logon tunnel, not cached-credential settings.

§Password change fails — expired password won't change off-network

Symptom: At the logon/lock screen off-network, the user cannot change an expired password ("Unable to update the password. The value provided … does not meet …" is a *policy* error; "no DC" is a *reachability* error).

1. **No DC path at the moment of change.** The AD password change must reach a DC. Bring the pre-logon

tunnel up **first** (device tunnel should already be up; SBL/PLAP the user connects first), then change the password. Confirm DC reachability with `PREVPN-Validation.ps1`.

1. **Clock skew.** Kerberos rejects password operations when device time is too far from the DC's

(default tolerance five minutes). Correct the device clock/time zone and retry.

1. **SSPR assumed to be enough (it isn't, for hybrid/domain-joined).** SSPR resets the **directory**

password in the cloud; with writeback it updates the on-prem directory — but the **local interactive change at the Windows lock screen still expects a DC path**. Use SSPR/TAP for the cloud-capable paths, but for a hybrid/domain-joined device's lock-screen change, the DC path via the tunnel is still required. Validate the real behavior (T2).

1. **Password policy rejection (not a network problem).** If the tunnel and DC are fine but the change

is refused on complexity/history/age, it is an AD **policy** rejection — the user must pick a compliant password; this is not a pre-logon-VPN issue.

§Entra-joined confusion — "do cloud-only devices need this?"

Symptom: Uncertainty about whether an Entra-joined device needs a pre-logon tunnel; or a pre-logon tunnel was deployed to cloud-only devices unnecessarily.

1. **Entra-joined devices authenticate against the cloud, not a DC.** A first sign-in works over

ordinary internet and the password lives in Entra — **no pre-logon tunnel is needed**. Cached **cloud** credentials (PRT/Web Account Manager) cover subsequent sign-ins.

1. **Confirm the join type before deploying anything.** `dsregcmd /status`: `AzureAdJoined=YES`,

`DomainJoined=NO`, `EnterpriseJoined=NO` = **Entra-joined**. `DomainJoined=YES` + `AzureAdJoined=YES` = **hybrid-joined** (needs a DC path). `PREVPN-Discovery.ps1` reports the mix.

1. **Remove over-deployment.** If a device tunnel/SBL was targeted at Entra-joined devices, exclude

them from the chosen approach's assignment — they don't need it and it only adds surface/support.

1. **Password path for Entra-joined:** use **SSPR** and **TAP**; there is no on-prem DC dependency.

§Pre-logon path not present — validation says no tunnel

Symptom: `PREVPN-Validation.ps1` reports no device tunnel and no SBL/PLAP provider on a device that should have one.

1. **Chosen approach's object not deployed/assigned.** Confirm `AOVPN – Device Tunnel – ProfileXML –

Windows (device tunnel) or Cisco Secure Client – Pre-Logon (SBL) – Win32` (SBL) exists and is assigned to the device's group — deploy/assign it via the **companion product**.

1. **Device not in the segmentation group.** Confirm the device/user is in `PILOT – Pre-Logon

First-Logon Users (or the prod group) so the companion product's assignment reaches it. Use PREVPN-Remediation.ps1` to add lacking devices/users, then sync.

1. **Not yet synced.** Force a sync and allow time for the companion product's profile/app to land,

then re-run validation.

§Script / reporting issues

1. `Connect-MgGraph` **fails or lacks scopes.** Install the SDK (`Install-Module Microsoft.Graph

-Scope CurrentUser) and consent to the scopes each script lists in its header. Read-only scripts need only .Read.All scopes. (PREVPN-Validation.ps1` needs no Graph — it runs locally.)

1. **Join type shows unknown.** `PREVPN-Validation.ps1` parses `dsregcmd /status`; run it in a normal

user context on the device. `PREVPN-Discovery.ps1` derives join type from Entra device `trustType` — a device missing from Entra shows unknown until it registers.

1. **Remediation "made no changes."** By design — it defaults to **-WhatIf**. Re-run with

`-WhatIf:$false` after reviewing the previewed actions. Remember it only adjusts **group membership** for the chosen approach; it never creates VPN profiles.

If first-logon or password-change still fails after the decision side is confirmed healthy (right approach for the join type, pre-logon path present, DC reachable, clock in sync), treat it as a **companion-product / network / vendor head-end issue** and continue with that product's troubleshooting and the **vendor's official documentation**.

Best Practices — Pre-Logon VPN: First-Logon & Off-Network Password Reset

Field-proven practices for the decision + scenario + validation layer. These keep the pre-logon decision defensible, the scenarios reliable, and the fleet free of unnecessary tunnels.

Decide by join type first

- **Start from the join-type inventory, always.** Run `PREVPN-Discovery.ps1` before deciding anything.

The first-logon / DC-reachability problem is a **hybrid/domain-joined** problem; deciding without the join-type mix leads to over-deploying to cloud-only devices.

- **Never deploy a pre-logon tunnel to Entra-joined (cloud-only) devices.** They authenticate against

Entra in the cloud; a first sign-in and password change work without an on-prem DC. Use SSPR/TAP for their cloud password paths instead.

- **Let the matrix pick the approach, and record why.** Device tunnel for the Enterprise/Education +

machine-cert + IKEv2 cohort (silent, no user action); SBL/PLAP where you already run Cisco Secure Client or can't meet that bar (user-initiated at the logon screen). Write the decision down.

Keep the three surfaces coherent

- **Reference the companion objects by their exact names.** `AOVPN – Device Tunnel – ProfileXML – Windows and Cisco Secure Client – Pre-Logon (SBL) – Win32` are created in the companion products — never rename or fork them here; this product only decides and validates.

- **Don't rebuild the tunnel in this layer.** If you find yourself editing a ProfileXML or the SBL

module, you're in the wrong product — do that work in the companion product so the build stays in one place.

- **Segment cohorts cleanly.** Use `PILOT – Pre-Logon First-Logon Users` and `PROD – Pre-Logon First-Logon Users` to stage validation and expansion; keep the pilot group assigned after expansion as a fast test cohort.

Validate the scenarios, not just the plumbing

- **Always run T1 and T2 end to end on the pilot.** A tunnel that "exists" is not proof — prove that a

never-logged-on user completes first logon off-network and that an **expired-password** user can change it at the lock screen. These two tests are the whole point.

• **Include an Entra-joined control in the pilot.** Confirm it signs in with **no** pre-logon path — it is your evidence that cloud-only devices don't need this deployment.

• **Confirm DC reachability separately.** The tunnel coming up and a DC being reachable through it are two different things — validate both (T3 and T4).

Certificates and SKU (device tunnel)

• **Verify the machine cert and SKU before promising a silent tunnel.** The device tunnel is silent only when a valid **LocalMachine\My** machine certificate exists and the device is **Enterprise/Education**. Confirm both (T5/R6) or route those devices to SBL/PLAP.

• **Watch machine-cert lifetime.** A lapsed machine cert silently breaks the pre-logon tunnel — monitor renewal in the companion product so devices never lose the DC path at logon.

SSPR is a complement, not a replacement

• **Enable SSPR/TAP for the cloud-capable paths**, but do not assume they remove the DC-path requirement for a **hybrid/domain-joined** device's lock-screen password change. Document exactly where SSPR helps (Entra-joined; directory reset via writeback) and where the DC path is still needed.

Break-glass, always ready

• **Keep a documented fallback for every stranded-user case:** local administrator recovery, a **Temporary Access Pass** for cloud-capable first sign-in, and a "bring it on-network once" path. Test the fallback (T8) before you need it.

• **Onboard the very first device on a known network when in doubt.** For a brand-new hire, a single first logon on a trusted network primes the cached credential and removes the off-network first-logon risk entirely — a pragmatic fallback when pre-logon VPN isn't ready.

Operate and monitor

• **Run** `PREVPN-Report.ps1` **on a schedule** to track pre-logon coverage of in-scope devices as the fleet changes, and to catch cloud-only devices that were targeted by mistake.

• **Remediate safely.** `PREVPN-Remediation.ps1` defaults to **-WhatIf** and only adjusts **group membership** for the chosen approach — it never creates VPN profiles. Preview first, then re-run with `-WhatIf:$false`.

• **Least-privilege admin.** Groups Administrator + Intune Administrator for the operator; reserve Global Administrator for tenant-wide changes only.

Deliverables Summary — Pre-Logon VPN: First-Logon & Off-Network Password Reset

This solution is a complete, self-guided **decision + scenario + validation** package for pre-logon VPN — it helps you **choose** the right approach (Always On VPN device tunnel vs. Cisco Secure Client SBL/PLAP) for your fleet, explains the **two hard scenarios** by device join type, and **validates** them end to end. It **references** the two deployment products by their object names but does not re-teach their builds. Below is everything included.

Documentation set (source/)

Document	What it gives you
`01-solution-overview.md`	The problem (no cached creds for first logon; expired password can't reach a DC), the two approaches, the decision matrix, why join type is the whole story, both scenarios, break-glass, and the Assess→Configure→Pilot→Validate→Expand model.
`02-requirements.md`	Every prerequisite with **Description / Why / Verify / Implement-if-missing** — a chosen/deployed approach (or the companion product), segmentation groups, join-type inventory, DC reachability, machine cert + SKU (device tunnel only), optional SSPR, break-glass accounts.
`03-implementation.md`	The ordered procedure: run readiness, **choose the approach via the decision matrix**, segment the pilot, confirm the chosen companion object, then **validate the two scenarios** — with Gates A–E framed around scenario validation. Includes the decision matrix as a table.
`05-testing.md`	The validation matrix (T1–T8) with explicit **PASS/FAIL** gates — the two decisive scenario tests (first-time off-network logon; off-network password change) plus pre-logon-path, DC-reachability, machine-cert, Entra-joined-control, coverage, and break-glass tests.
`06-troubleshooting.md`	Failure-by-symptom fixes: first logon still fails, password change fails, Entra-joined confusion, pre-logon path not present, script/reporting issues.
`07-best-practices.md`	Field-proven practices for deciding by join type, keeping the three surfaces coherent, validating scenarios not plumbing, certs/SKU, SSPR as a complement, and break-glass.
`executive-summary.md`	One-page business-outcome summary for stakeholders.
`global-requirements-network.md`	Category-level Network prerequisites (shared across Network solutions).

PowerShell scripts (scripts/)

Discovery/report use the **Microsoft Graph PowerShell SDK** (read-only); validation runs **locally on a device** (no Graph); remediation and scope-targeting **default to** `-WhatIf` **(preview)** and change nothing until you re-run with `-WhatIf:$false`. Each script's header documents its exact purpose, run command, and required Graph scopes.

Script	Purpose	Mode
`PREVPN-Discovery.ps1`	Fleet readiness: reports device **join types** (hybrid / domain-joined / Entra-joined), presence of a pre-logon capability (a device-tunnel ProfileXML profile **or** the SBL Win32 app), and pilot/prod group membership; recommends which approach fits the fleet.	Read-only
`PREVPN-Validation.ps1`	**On this device:** detects join type (`dsregcmd /status`), whether a pre-logon tunnel exists (device tunnel via `Get-VpnConnection -AllUserConnection` **or** PLAP `vpnplap64.dll`), the machine cert (LocalMachine\My), and a best-effort **DC reachability** check. PASS = pre-logon capability present AND a DC reachable.	Read-only (local)
`PREVPN-Report.ps1`	Fleet readiness roll-up — counts by join type and by pre-logon capability present/absent, with a **coverage %** of in-scope devices that have a pre-logon path.	Read-only
`PREVPN-Remediation.ps1`	Safely adds devices/users lacking a pre-logon path to the pilot group for the chosen approach and prints guidance. **Does NOT create VPN profiles** (that's the companion products).	Write, **default** `-WhatIf`
`PREVPN-Set-ScopeTargeting.ps1`	Creates/populates `PILOT – Pre-Logon First-Logon Users` (and optional PROD) as assignable security groups. Idempotent.	Write, **default** `-WhatIf`

The consistent object naming convention

Every artifact is named to a convention and reused verbatim across docs, tests, troubleshooting, and scripts (so the scripts find objects deterministically):

• Segmentation groups: `PILOT – Pre-Logon First-Logon Users`, `PROD – Pre-Logon First-Logon Users`

• Referenced (created in companion products): the device-tunnel profile `AOVPN – Device Tunnel – ProfileXML – Windows and the SBL app Cisco Secure Client – Pre-Logon (SBL) – Win32`

Scope boundary (what is deferred)

This is the **decision + scenario + validation** layer. The **step-by-step device-tunnel and SBL/PLAP builds live in their companion products** and, with IKEv2/ProfileXML and Cisco head-end internals, are **deferred to those products and the vendor's official documentation**. This solution chooses between the two approaches, explains and validates the two scenarios by join type, and reports fleet readiness — it consumes the companion products' objects rather than recreating them.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

PREVPN-Discovery.ps1

Purpose: Read-only readiness sweep for the pre-logon VPN decision. Reports how the fleet's Windows devices are joined (Entra hybrid-joined vs Entra-joined vs domain-joined) from Entra device trustType, whether a pre-logon capability is deployed (a device-tunnel ProfileXML profile OR the Cisco Secure Client SBL Win32 app), and the pilot/production segmentation-group membership. Ends with a recommendation of which approach fits the fleet. Makes NO changes and creates NO VPN profiles (deployment is done in the companion products). HOW TO RUN powershell -ExecutionPolicy Bypass -File .\PREVPN-Discovery.ps1 Required Microsoft Graph scopes (delegated, read-only): Device.Read.All DeviceManagementApps.Read.All DeviceManagementConfiguration.Read.All Group.Read.All Directory.Read.All EXPECTED RESULTS A readiness table: join-type counts (hybrid / Entra-joined / domain-joined / unknown), whether the referenced device-tunnel profile and the SBL app are Found/Missing, pilot/prod group counts, and a recommended approach. Hybrid/domain-joined devices are the ones the two scenarios apply to; Entra-joined largely avoid them. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. Vendor internals deferred to vendor docs. >

How to run: powershell -ExecutionPolicy Bypass -File .\PREVPN-Discovery.ps1 Required Microsoft Graph scopes (delegated, read-only): Device.Read.All DeviceManagementApps.Read.All DeviceManagementConfiguration.Read.All Group.Read.All Directory.Read.All EXPECTED RESULTS A readiness table: join-type counts (hybrid / Entra-joined / domain-joined / unknown), whether the referenced device-tunnel profile and the SBL app are Found/Missing, pilot/prod group counts, and a recommended approach. Hybrid/domain-joined devices are the ones the two scenarios apply to; Entra-joined largely avoid them. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. Vendor internals deferred to vendor docs. >

PREVPN-Remediation.ps1

Purpose: Safe remediation for the pre-logon VPN decision. Finds IN-SCOPE Windows devices (Entra hybrid-joined / domain-joined) that are NOT yet members of the segmentation group for the chosen approach, and adds them so the companion product's pre-logon assignment reaches them. It does NOT create VPN profiles or apps — that is the job of the companion device-tunnel / SBL-PLAP products. Defaults to preview only (-WhatIf): it PRINTS what it would do and changes nothing until you re-run with -WhatIf:$false. Entra-joined (cloud-only) devices are skipped by design (they do not need a pre-logon tunnel). HOW TO RUN Preview (default, no changes): powershell -ExecutionPolicy Bypass -File .\PREVPN-Remediation.ps1 Apply changes: powershell -ExecutionPolicy Bypass -File .\PREVPN-Remediation.ps1 -WhatIf:$false Required Microsoft Graph scopes (delegated): Device.Read.All Group.ReadWrite.All GroupMember.ReadWrite.All Directory.Read.All EXPECTED RESULTS A plan listing each in-scope device to add to the target group. In preview every line is prefixed [WHATIF]. With -WhatIf:$false each add prints [DONE]/[FAILED]. After applying, deploy (or confirm) the chosen approach via its companion product, then re-run PREVPN-Report.ps1. .NOTES White-label — the repeatable way. Write actions gated behind -WhatIf (default ON). No VPN profiles created here. Vendor internals deferred to vendor docs. >

How to run: Preview (default, no changes): powershell -ExecutionPolicy Bypass -File .\PREVPN-Remediation.ps1 Apply changes: powershell -ExecutionPolicy Bypass -File

.\PREVPN-Remediation.ps1 -WhatIf:$false Required Microsoft Graph scopes (delegated): Device.Read.All Group.ReadWrite.All GroupMember.ReadWrite.All Directory.Read.All EXPECTED RESULTS A plan listing each in-scope device to add to the target group. In preview every line is prefixed [WHATIF]. With -WhatIf:$false each add prints [DONE]/[FAILED]. After applying, deploy (or confirm) the chosen approach via its companion product, then re-run PREVPN-Report.ps1. .NOTES White-label — the repeatable way. Write actions gated behind -WhatIf (default ON). No VPN profiles created here. Vendor internals deferred to vendor docs. >

PREVPN-Report.ps1

Purpose: Read-only fleet roll-up for the pre-logon VPN decision. Counts Windows devices by join type (Entra hybrid-joined / domain-joined / Entra-joined / registered / unknown) and estimates pre-logon-path coverage for the IN-SCOPE (hybrid + domain-joined) devices by whether the referenced pre-logon objects are deployed and assigned — the device-tunnel ProfileXML profile and/or the SBL Win32 app. Prints a coverage percentage and an optional CSV. Makes NO changes. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\PREVPN-Report.ps1 Optional: -CsvPath .\prevpn-readiness.csv to export the per-join-type rows. Required Microsoft Graph scopes (delegated, read-only): Device.Read.All DeviceManagementApps.Read.All DeviceManagementConfiguration.Read.All EXPECTED RESULTS A join-type table, a pre-logon-capability table (device tunnel / SBL present + assigned), and a coverage % = share of in-scope devices that have a deployed+assigned pre-logon path. Target: pilot 100% of in-scope devices; expansion >= your threshold (e.g. 95%). Entra-joined devices are reported separately and excluded from the in-scope denominator. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. Vendor internals deferred to vendor docs. >

How to run: powershell -ExecutionPolicy Bypass -File .\PREVPN-Report.ps1 Optional: -CsvPath .\prevpn-readiness.csv to export the per-join-type rows. Required Microsoft Graph scopes (delegated, read-only): Device.Read.All DeviceManagementApps.Read.All DeviceManagementConfiguration.Read.All EXPECTED RESULTS A join-type table, a pre-logon-capability table (device tunnel / SBL present + assigned), and a coverage % = share of in-scope devices that have a deployed+assigned pre-logon path. Target: pilot 100% of in-scope devices; expansion >= your threshold (e.g. 95%). Entra-joined devices are reported separately and excluded from the in-scope denominator. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. Vendor internals deferred to vendor docs. >

PREVPN-Set-ScopeTargeting.ps1

Purpose: Creates the segmentation target groups for the pre-logon VPN decision — 'PILOT – Pre-Logon First-Logon Users' and (optionally) 'PROD – Pre-Logon First-Logon Users' — as assignable security groups, and adds the members you supply (by user principal name). Idempotent: if a group already exists it is reused, and members already present are skipped. Defaults to preview only (-WhatIf); re-run with -WhatIf:$false to actually create/populate. These groups are the cohorts you validate the two scenarios against; the chosen approach's VPN objects are then assigned to them in the COMPANION product. HOW TO RUN Preview: powershell -ExecutionPolicy Bypass -File .\PREVPN-Set-ScopeTargeting.ps1 -PilotMembers user1@your-domain.com,user2@your-domain.com Apply: powershell -ExecutionPolicy Bypass -File .\PREVPN-Set-ScopeTargeting.ps1 -PilotMembers user1@your-domain.com -WhatIf:$false Required Microsoft Graph scopes (delegated): Group.ReadWrite.All GroupMember.ReadWrite.All User.Read.All EXPECTED RESULTS The pilot group (and prod group if -CreateProd) exist and contain the listed members. In preview each create/add is prefixed [WHATIF]; with -WhatIf:$false each prints [DONE]/[FAILED]. These exact group names are the cohorts referenced in 03-implementation.md. .NOTES White-label — the repeatable way. Write actions gated behind -WhatIf (default ON). No VPN

profiles created here. Vendor internals deferred to vendor docs. >

How to run: Preview: powershell -ExecutionPolicy Bypass -File .\PREVPN-Set-ScopeTargeting.ps1 -PilotMembers user1@your-domain.com,user2@your-domain.com Apply: powershell -ExecutionPolicy Bypass -File .\PREVPN-Set-ScopeTargeting.ps1 -PilotMembers user1@your-domain.com -WhatIf:$false Required Microsoft Graph scopes (delegated): Group.ReadWrite.All GroupMember.ReadWrite.All User.Read.All EXPECTED RESULTS The pilot group (and prod group if -CreateProd) exist and contain the listed members. In preview each create/add is prefixed [WHATIF]; with -WhatIf:$false each prints [DONE]/[FAILED]. These exact group names are the cohorts referenced in 03-implementation.md. .NOTES White-label — the repeatable way. Write actions gated behind -WhatIf (default ON). No VPN profiles created here. Vendor internals deferred to vendor docs. >

PREVPN-Validation.ps1

Purpose: Local, read-only validation run ON a pilot Windows device. Detects the device join type (dsregcmd /status), whether a pre-logon tunnel exists (an Always On VPN DEVICE tunnel via Get-VpnConnection -AllUserConnection, OR the Cisco Secure Client Start Before Logon credential provider vpnplap64.dll), whether a machine/device certificate is present (LocalMachine\My), and a best-effort domain-controller reachability check. Prints a clear PASS/FAIL so a pilot device can be counted as pre-logon-ready. Makes NO changes. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\PREVPN-Validation.ps1 Optional: -DcFqdn dc01.your-domain.com check reachability to a specific DC (port 389) -RequireCertificate require a LocalMachine\My cert (device-tunnel deployments) No Microsoft Graph scopes needed — this inspects the local device only. EXPECTED RESULTS A checklist: join type (hybrid / domain-joined / Entra-joined), pre-logon path present (device tunnel or PLAP), machine cert present (if checked), and a DC reachable. Overall PASS when a pre-logon capability is present AND a domain controller is reachable through it. On an Entra-joined (cloud-only) device the script reports that no pre-logon tunnel is needed. .NOTES White-label — the repeatable way. Read-only local inspection. Vendor internals deferred to vendor docs. >

How to run: powershell -ExecutionPolicy Bypass -File .\PREVPN-Validation.ps1 Optional: -DcFqdn dc01.your-domain.com check reachability to a specific DC (port 389) -RequireCertificate require a LocalMachine\My cert (device-tunnel deployments) No Microsoft Graph scopes needed — this inspects the local device only. EXPECTED RESULTS A checklist: join type (hybrid / domain-joined / Entra-joined), pre-logon path present (device tunnel or PLAP), machine cert present (if checked), and a DC reachable. Overall PASS when a pre-logon capability is present AND a domain controller is reachable through it. On an Entra-joined (cloud-only) device the script reports that no pre-logon tunnel is needed. .NOTES White-label — the repeatable way. Read-only local inspection. Vendor internals deferred to vendor docs. >